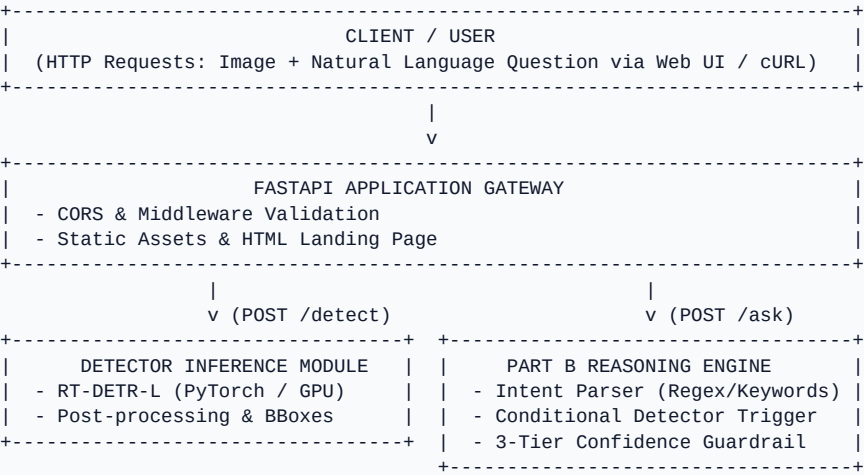


System Design & Architecture Specification

END-TO-END CV + REASONING PIPELINE ARCHITECTURE

1. High-Level Architecture Diagram



2. Core Subsystem Responsibilities

- **Dataset Pipeline:** Range-request streaming HTTP downloader + single-class deduplication filter.
- **Model Inference Engine:** RT-DETR-L Transformer with AIFI (Intra-Scale Feature Interaction) and CCFM (Cross-Scale Feature Fusion).
- **Reasoning Layer:** Deterministic intent router handling COUNT, PRESENCE, LIST, SPATIAL, and MOST_COMMON queries.
- **Guardrail Enforcement:** Prevents hallucination by truncating low-confidence detections (<0.25).